

# CURRICULUM VITAE

**Ivan Loseu<sup>1</sup>.**

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## 1. CONTACT INFORMATION.

*E-mail:* ivan.loseu@gmail.com

## 2. PERSONAL INFORMATION

*Born:* October, 18, 1981, Minsk, Republic of Belarus, USSR.

*Marital status:* Single.

*Citizenship:* Belarus, US.

Webpage.

## 3. EDUCATION

*1999-2004:* M.SC., Belarusian State University, Department of Applied Mathematics and Computer Science, diploma cum laud.

*2004-2007:* graduate student, Moscow State University, Department of Mechanics and Mathematics, Chair of Higher Algebra.

*2004-2007:* graduate student, Independent University of Moscow.

I obtained my PhD degree in October, 2007, at the Moscow State University.

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<sup>1</sup>"Loseu" is the transliteration of my family name from Belarusian to English. This variant appears in my passport and is the only version used officially. Usually I use the transliteration from Russian, which is "Losev". The reason is that people who do not know Russian or Belarusian spell the Belarusian variant incorrectly.

*Thesis advisor:* Prof. E.B. Vinberg.

*Thesis title:* Classification of some coisotropic actions of algebraic groups.

#### 4. EMPLOYMENT

*Starting July 2019:* Full Professor, Yale University, Department of Mathematics.

*July 2018-June 2019:* Full Professor, University of Toronto, Department of Mathematical and Computational Sciences, Mississauga campus.

*July 2015-June 2018:* Full Professor, Northeastern University, Department of Mathematics.

*Sep. 2011-June 2015:* Associate Professor, Northeastern University, Department of Mathematics.

*Jul.2008- Jun. 2011:* CLE Moore instructor, Massachusetts Institute of Technology, Department of Mathematics.

*Nov. 2007- May 2008:* Engineer-programmer, Belarusian State University, Department of Applied Mathematics and Computer Science.

#### 5. VISITING POSITIONS

*March 2017- December 2019:* Research fellow and (starting July 2017) Chief Research fellow, International laboratory of Representation theory and Mathematical Physics, NRU-HSE, Moscow, Russia.

*March-May 2018:* Research Professor, MSRI.

#### 6. RESEARCH INTERESTS

In general: Representation theory and its connections to Algebraic and Symplectic geometry.

Currently: Harish-Chandra modules over quantizations, quantum groups at roots of unity, the “double affine” representation theory.

#### 7. PUBLICATION LIST

##### **Publications in refereed journals.**

[1] I.V. Losev. *Coisotropic representations of reductive groups*. Trudy Mosc. Mat. Ob-va, 66(2005), p. 157-181 (in Russian). English translation in: Trans. Moscow Math. Soc. (2005), 143-168.

[2] I.V. Losev. *Symplectic slices for reductive groups*. Mat. Sbornik 197(2006), N2, p. 75-86 (in Russian). English translation in: Sbornik Math. 197(2006), N2, 213-224.

[3] I.V. Losev. *On complex weakly commutative homogeneous spaces*. Trudy Mosc. Mat. Ob-va, 67(2006), 228-255 (in Russian). English translation in: Trans. Moscow Math. Soc. (2006), 199-223.

[4] I.V. Losev. *Computation of the Cartan spaces of affine homogeneous spaces*. Mat. Sbornik, 198(2007), 83-108 (in Russian). English translation in: Sbornik Math. 198(2007), no 10, 31-56. arXiv:math.AG/0606101v2.

[5] I.V. Losev. *Classification of weakly commutative complex homogeneous spaces*. Usp. Mat. Nauk 62(2007) N2, 181-182 (in Russian).

[6] I.V. Losev. *Combinatorial invariants of algebraic Hamiltonian actions*. Moscow Math. J. 8(2008), 493-519. arXiv:math.AG/0701823.

- [7] I.V. Losev. *Uniqueness property for spherical homogeneous spaces*. Duke Math J., 147(2009), n.2, 315-343. arXiv:math.AG/0703543.
- [8] I.V. Losev. *Proof of the Knop conjecture*. Ann. Inst. Fourier, 59(2009), n.3, 1105-1134. arXiv:math.AG/0612561.
- [9] I.V. Losev. *Demazure embeddings are smooth*. Int. Math. Res. Not, 14(2009), 2588-2596. arXiv:0704.3698.
- [10] I.V. Losev. *Lifting central invariants for quantized Hamiltonian actions*. Moscow Math J. 9(2009), 359-369. arXiv:0708.0630.
- [11] I.V. Losev. *Computation of weight lattices of  $G$ -varieties*. J. Math Sci 161(2009), N1, 70-96. arXiv:0709.0667v1.
- [12] I. Losev. *Classification of multiplicity free Hamiltonian actions of complex tori on Stein manifolds*. J. Sympl. Geom 7(2009), N3, 295-310. arXiv:0706.0632
- [13] I.V. Losev. *Algebraic Hamiltonian actions*, Math. Z. 263(2009), 685-723. arXiv:math.AG/0601023.
- [14] I.V. Losev. *Embeddings of homogeneous spaces into irreducible modules*. J. Algebra 322 (2009), 2621-2630, arXiv:math.RT/0606387.
- [15] I.V. Losev. *On fibers of algebraic invariant moment maps*. Transformation Groups, 14(2009), 887-930. arXiv:math. AG/0703296.
- [16] I.V. Losev. *Quantized symplectic actions and  $W$ -algebras*. J. Amer. Math. Soc. 23(2010), 35-59. arXiv:0707.3108.
- [17] I.V. Losev. *Computation of Weyl groups of  $G$ -varieties*. Representation Theory (electronic) 14(2010), 9-69. arXiv:0612559.
- [18] I. Losev. *On invariants of a set of elements of a semisimple Lie algebra*. J. Lie Theory, 20(2010), 17-30. arXiv:math.RT/0512538.
- [19] I. Losev. An appendix to: P. Etingof, T. Schedler, *Poisson traces and  $D$ -modules on Poisson varieties*, GAFA 20(2010), 958-987. arXiv:0908.3868.
- [20] I. Losev. *1-dimensional representations and parabolic induction for  $W$ -algebras*. Adv. Math. 226(2011), 6, 4841-4883. arXiv:0906.0157.
- [21] I.V. Losev. *Finite dimensional representations of  $W$ -algebras*. Duke Math J. 159(2011), n.1, 99-143. arXiv:0807.1023.
- [22] I. Losev. *On the structure of the category  $\mathcal{O}$  for  $W$ -algebras*. Séminaires et Congrès 24(2012), 351-368. arXiv:0812.1584.
- [23] I. Losev, *Completions of symplectic reflection algebras*. Selecta Math., 18(2012), N1, 179-251. arXiv:1001.0239.
- [24] I. Losev, *Primitive ideals in  $W$ -algebras of type  $A$* . J. Algebra, 359 (2012), 80-88. arXiv:1108.4171.
- [25] I. Losev, *Isomorphisms of quantizations via quantization of resolutions*. Adv. Math. 231(2012), 1216-1270. arXiv:1010.3182.
- [26] I. Losev, *On isomorphisms of certain functors for Cherednik algebras*. Repres. Theory, 17 (2013), 247-262. arXiv:1011.0211.
- [27] I. Losev, *Highest weight  $\mathfrak{sl}_2$ -categorifications I: crystals*. Math. Z. 274(2013), 1231-1247. arXiv:1201.4493.
- [28] I. Losev, A. Tsymbaliuk, *Infinitesimal Cherednik algebras as  $W$ -algebras*. arXiv:1305.6873. Transform. Groups. 19 (2014), 495-526.
- [29] I. Gordon, I. Losev, *On category  $\mathcal{O}$  for cyclotomic rational Cherednik algebras*. J. Eur. Math. Soc. 16 (2014), 1017-1079.

- [30] I. Losev, V. Ostrik, *Classification of finite dimensional irreducible modules over  $W$ -algebras*. arXiv:1202.6097. Compos. Math. 150(2014), N6, 1024-1076.
- [31] I. Losev, *On Procesi bundles*. arXiv:1303.4617. Math. Ann. 359(2014), N3, 729-744.
- [32] I. Losev. *Finite dimensional quotients of Hecke algebras*. arXiv:1407.6375. Algebra and Number theory, 9(2015), 493-502.
- [33] I. Losev, B. Webster, *On uniqueness of tensor products of irreducible categorifications*. arXiv:1303.1336. Selecta Math. 21(2015), N2, 345-377.
- [34] P. Etingof, E. Gorsky, I. Losev, *Representations of Cherednik algebras with minimal support and torus knots*. arXiv:1304.3412. Adv. Math. 277 (2015), 124-180.
- [35] I. Losev, *Dimensions of irreducible modules over  $W$ -algebras and Goldie ranks*. arXiv:1209.1083. Invent. Math. 200 (2015), N3, 849-923.
- [36] I. Losev. *Abelian localization for cyclotomic Cherednik algebras*. Int Math Res Notices (2015) vol. 2015, 8860-8873. arXiv:1402.0224.
- [37] I. Losev, *Highest weight  $\mathfrak{sl}_2$ -categorifications II: structure theory*. Trans. Amer. Math. Soc. 367 (2015) 8383-8419. arXiv:1203.5545.
- [38] I. Losev, *Proof of Varagnolo-Vasserot conjecture on cyclotomic categories  $\mathcal{O}$* . arXiv:1305.4894. Selecta Math. 22(2016), 631-668.
- [39] I. Losev. *Derived equivalences for Rational Cherednik algebras*. arXiv:1406.7502. Duke Math J. 166(2017), N1, 27-73.
- [40] I. Losev. *Bernstein inequality and holonomic modules* (with a joint appendix by I. Losev and P. Etingof). arXiv:1501.01260. Adv. Math. 308 (2017), 941-963.
- [41] I. Losev. An appendix to *Quantizations of conical symplectic resolutions II: category  $\mathcal{O}$  and symplectic duality* by T. Braden, A. Licata, N. Proudfoot, B. Webster. arXiv:1407.0964. Astérisque 384 (2016), 75-179.
- [42] I. Losev. *On categories  $\mathcal{O}$  for quantized symplectic resolutions*. arXiv:1502.00595. Compos. Math. 153 (2017), N12, 2445-2481.
- [43] J. Brundan, I. Losev, B. Webster, *Graded tensor product categorifications and the super Kazhdan-Lusztig conjecture*. arXiv:1310.0349. Int. Math. Res. Notices. (2017), vol. 2017 n. 20, 6329-6410.
- [44] I. Losev. *Wall-crossing functors for quantized symplectic resolutions: perversity and partial Ringel dualities*. PAMQ, 13 (2017), n.2, 247-289.
- [45] I. Losev. *Quantizations of regular functions on nilpotent orbits*. Bull. Inst. Math. Acad. Sin. 13 (2018), n.2, 199-225.
- [46] I. Losev, S. Shelley-Abrahamson. *On Refined Filtration By Supports for Rational Cherednik Categories  $\mathcal{O}$* . Selecta Math. 24 (2018), 1729-1804.
- [47] I. Losev. *Representation theory of quantized Gieseker varieties, I*. Lie groups, Geometry, and Representation theory, Progress in Mathematics, 326 (2018), 273-314.
- [48] R. Bezrukavnikov, I. Losev, *On dimension growth of modular irreducible representations of semisimple Lie algebras*. Lie groups, Geometry, and Representation theory, Progress in Mathematics, 326 (2018), 59-90.
- [49] I. Losev. *Cacti and cells*. J. Eur. Math. Soc. 21 (2019), 1729-1750.
- [50] I. Losev, *Derived equivalences for Symplectic reflection algebras*. Int. Math. Res. Not. IMRN 2021, no. 1, 444-474.
- [51] I. Losev. *Supports of simple modules in cyclotomic Cherednik categories  $\mathcal{O}$* . Adv. Math. 377(2021), article 107491.
- [52] R. Bezrukavnikov, I. Losev, *Etingof conjecture for quantized quiver varieties*. Invent. Math. 223 (2021), 1097-1226.

- [53] I. Losev, I. Panin, *Goldie ranks of primitive ideals and indexes of equivariant Azumaya algebras*. Moscow Math. J. 21 (2021), N2, 383-399.
- [54] P. Etingof, V. Krylov, I. Losev, J. Simental, *Representations with minimal support for quantized Gieseker varieties*. Math. Z. 298 (2021), N3, 1593-1621.
- [55] I. Losev, *Harish-Chandra bimodules over quantized symplectic singularities*. Transform. Groups. 26 (2021), N2, 565-600.
- [56] I. Losev. *Deformations of symplectic singularities and Orbit method for semisimple Lie algebras*. Selecta Math, 28 (2022), N2, paper N30, 52 pages.
- [57] I. Losev. *Totally aspherical parameters for Cherednik algebras*. “Representation theory and Algebraic geometry”, 37-56, Trends in Mathematics, Springer 2022.
- [58] I. Losev, *On inductive construction of Procesi bundles*. European Journal of Mathematics, 8 (2022), 1006–1035.
- [59] R. Bezrukavnikov, I. Losev, *Dimensions of modular irreducible representations of semisimple Lie algebras*. arXiv:2005.10030. J. Amer. Math. Soc. 36 (2023), 1235-1304.
- [60] I. Halacheva, A. Licata, I. Losev, O. Yakobi, *Categorical braid group actions and cactus groups*. arXiv:2101.05931. Adv. Math. 429 (2023), 109190.
- [61] P. Boixeda Alvarez, I. Losev, *Affine Springer Fibers, Procesi bundles, and Cherednik algebras*. arXiv:2104.09543. Duke Math J. 173 (2024), 806-872.
- [62] I. Losev, *On modular categories  $O$  for quantized symplectic resolutions*. arXiv:1712.07726. Pure and Applied Mathematics Quarterly 21 (2025), N1, 415-494.
- [63] I. Losev, *Canonical filtrations on Harish-Chandra modules*. arXiv:2309.12007. Representation theory 29 (2025), 136-150.

### Accepted.

- [64] I. Losev, L. Mason-Brown, D. Matvieievskiy, *Unipotent Ideals and Harish-Chandra Bimodules*. arXiv:2108.03453. Accepted to Annals of Math. Studies.

### Preprints.

- [65] I.V. Losev. *The Kempf-Ness theorem and Invariant theory*. Preprint (2006), arXiv:math.AG/0605756.
- [66] I.V. Losev. *Computation of combinatorial invariants of  $G$ -varieties*. Preprint (2006), available at: [www.moebiuscontest.ru/files/2006/losev.pdf](http://www.moebiuscontest.ru/files/2006/losev.pdf) (in Russian).<sup>2</sup>
- [67] I. Losev, *Quantizations of nilpotent orbits vs 1-dimensional representations of  $W$ -algebras*. arXiv:1004.1669.
- [68] I. Losev, *Towards multiplicities for cyclotomic rational Cherednik algebras*. arXiv:1207.1299.<sup>3</sup>
- [69] I. Losev. *Etingof conjecture for quantized quiver varieties II: affine quivers*. arXiv:1405.4998.<sup>4</sup>
- [70] B. Elias, I. Losev, *Modular representation theory in type  $A$  via Soergel bimodules*. arXiv:1701.00560.
- [71] I. Losev, *Localization theorems for quantized symplectic resolutions*. arXiv:2103.11193.
- [72] I. Losev, *Almost commuting varieties for symplectic Lie algebras*. arXiv:2104.11000.
- [74] I. Losev, *On modular Soergel bimodules, Harish-Chandra bimodules, and category  $O$* . arXiv:2302.05782.
- [75] I. Losev, S. Yu, *On Harish-Chandra modules over quantizations of nilpotent orbits*. arXiv:2309.11191.

<sup>2</sup>Contains the results of [11],[15],[17] together with some additional results

<sup>3</sup>This preprint is replaced by arXiv:1305.4894

<sup>4</sup>This preprint is superseded by arXiv:1502.00595, arXiv:1604.06678, arXiv:1611.08470.

[76] I. Losev, *Quantum category  $O$  vs affine Hecke category*. arXiv:2310.03153.

[77] R. Bezrukavnikov, I. Losev, *Categorical Heisenberg action I: rational Cherednik algebras*. arXiv:2408.02485.

### Expository texts.

[a] I. Losev. *Quantized Hamiltonian actions of reductive groups and their applications*. In "Fundamental mathematics in the work of young scientists". Moscow, MCCME, 2009, p.64-80.

[b] I. Losev. *Uniqueness properties for spherical varieties*. <sup>5</sup> Les cours de CIRM, Année 2010, [http://ccirm.cedram.org/ccirm-bin/fget?id=CCIRM\\_2010](http://ccirm.cedram.org/ccirm-bin/fget?id=CCIRM_2010), 113-120. arXiv:0904.2937.

[c] I. Losev, *Finite  $W$ -algebras*. Proceedings of the International Congress of Mathematicians Hyderabad, India, 2010, p. 1281-1307. arXiv:1003:5811.

[d] I. Losev, *Representations of general linear groups and categorical actions of Kac-Moody algebras*. arXiv:1209.1067.

[e] I. Losev, *Procesi bundles and Symplectic reflection algebras*. arXiv:1501.00643. Algebraic and Analytic Microlocal Analysis, Springer Proceedings in Mathematics and Statistics, 269 (2018), 3-62.

[f] I. Losev, *Rational Cherednik algebras and categorification*. arXiv:1509.08550. Contemp. Math. 683, "Categorification and Higher Representation theory", A. Beliakova, A. Lauda, eds, 1-41.

## 8. GRANTS, AWARDS, AND DISTINCTIONS

- 1) 2004-2007: RFBR grant 05-01-00988.
- 2) *The 10th Moebius contest, Moscow, 2006*: the first prize<sup>6</sup>, <http://www.moebiuscontest.ru/history.php?l=0&part=11>.
- 3) 2009-2013: NSF grant DMS-0900907, \$137,751.
- 4) 2010: Invited sectional speaker for ICM 2010.
- 5) 2012-2016: NSF grant DMS-1161584, \$129,999.
- 6) *Fall 2014- Fall 2016*: Sloan fellowship, \$50,000.
- 7) 2015, NSF conference grant DMS-1507869, \$40,000.
- 8) 2015-2018, NSF grant DMS-1501558, \$309,060.
- 9) 08/2017-09/2019: Co-PI on the NSF RTG grant, DMS-1645877,  $\approx$ \$ 2,200,000.
- 10) 2017, NSF conference grant DMS-1744157, \$12,500.
- 11) AMS fellow, Class of 2018.
- 12) 2020-2026, NSF grant DMS-2001139, \$560,000.
- 13) 2021-2022, NSF conference grant DMS-2200713, \$49,000.
- 14) 2024, Simons fellowship.
- 15) 2025-2027, NSF grant DMS-2501558, \$260,000.

## 9. TEACHING

1) Fall 2005, Independent University of Moscow: A course "Moment map", lecturer. The program and lecture notes (in Russian) are available at:

[www.iu.mccme.ru/f05/momentum.html](http://www.iu.mccme.ru/f05/momentum.html)

2) Fall 2005, Spring 2006, Fall 2006: Independent University of Moscow: Algebra seminars for first year students, teaching assistant.

<sup>5</sup>Expanded lecture notes of a mini-course given at CIRM, Luminy, in April 2009

<sup>6</sup>awarded for the preprint [29]

- 3) Fall 2006, Independent University of Moscow: A course "Lie groups and Lie algebras", lecturer. The program and exercise sheets (in Russian) are available at:  
[www.iuum.mccme.ru/f06/lie.html](http://www.iuum.mccme.ru/f06/lie.html)
- 4) Fall 2007, Spring 2008, Belarusian State University, Minsk, Belarus: A course "Young tableaux", lecturer.
- 5) Spring 2008, Belarusian State University: A course "Advanced linear algebra", lecturer.
- 6) Spring 2008, Number theory and Geometry lectures for the Belarusian IMO team.
- 7) Fall 2008, MIT: section leader for single variable calculus course (18.01) with Prof. B. Brubaker.
- 8) Spring 2009, MIT: section leader for differential equations course (18.03) with Prof. D. Jerison.
- 9) August 2009, Belarusian State University: A crash-course "Hamiltonian mechanics and representations of quivers", lecturer.
- 10) Fall 2009, MIT: section leader for 18.01 with Prof. B. Brubaker.
- 11) Spring 2010, MIT: section leader for Project lab in math (18.821) with Prof. D. Jerison.
- 12) August 2010, Belarusian State University: A crash-course "Quadratic forms over  $\mathbb{Q}$ ".
- 13) Fall 2010, MIT: section leader for 18.01 with Prof. P. Seidel.
- 14) Spring 2011, MIT: section leader for 18.03 with Prof. D. Jerison.
- 15) July 2011, summer school "Modern Mathematics", Dubna, Russia: A crash course "Representations of groups and the Burnside theorem", lecturer.
- 16) December 2011, Higher School of Economics, Moscow Russia: A crash course "Calogero-Moser systems and Rational Cherednik algebras", 4 hours.
- 17) Spring 2012, Northeastern University. MATH 4555, Complex variables. Lecturer.
- 18) July 2012, summer school "Modern Mathematics", Dubna, Russia: a 5 hour crash course "Quantum groups, knots, and Jones polynomial", lecturer.
- 19) Fall, 2012, Northeastern University. MATH 1242, Calculus II, Lecturer.
- 20) Fall, 2012, Northeastern University. MATH 7364, Topics in Representation Theory: Symplectic Reflection algebras.
- 21) Fall, 2013, Northeastern University. MATH 1341, Calculus I for Science and Engineering, lecturer and course coordinator.
- 22) May 2014, Higher School of Economics, Moscow, Russia. Crash course "Representations of quivers and deformed preprojective algebras".
- 23) Fall 2014, Northeastern University. MATH 1341, Calculus I for Science and Engineering, lecturer.
- 24) Fall 2014, Northeastern University. MATH 7322, Geometry 3 (Symplectic geometry).
- 25) July 2015, summer school "Modern Mathematics", Dubna, Russia: a 5 hour crash course "Representation theory of the symmetric groups as a first step towards categorical actions of Lie algebras".
- 26) Fall 2015, Northeastern University. MATH 7313, Representation theory (modern introduction).
- 27) December 2015, Higher School of Economics, Moscow. A crash-course "Representation theory of symmetric groups in positive characteristic and Kac-Moody algebra actions on categories". 4.5 hours, aimed at undergraduate and graduate students.
- 28) December 2015, Belarusian State University, Minsk. A talk "Equipartite polyhedra and tensor products" aimed at high school students, 1 hour.

29) July 2016, Summer school “Algebra and Geometry”, Yaroslavl, Russia. Lecture series “Hilbert schemes and Combinatorics” aimed at undergraduate and graduate students, 4.5 hours.

30) August 2016, A crash-course “Quadratic forms over  $\mathbb{Q}$ ” aimed at undergraduate students, Belarusian State University, Minsk, Belarus, 6 hours.

31) Fall 2016, MATH 1341, Calculus 1 for Science and Engineering, lecturer and course coordinator.

32) March 2017, A crash-course “Around representations of quivers” aimed at undergraduate and graduate students, Chebyshev laboratory, St. Petersburg, Russia, 8 hours.

33) April 2017, A crash-course “Representations of Rational Cherednik algebras”, HSE, Moscow, Russia, around 17 hours.

34) July 2017, Lecture series “Representations of groups and the Burnside theorem” aimed at high school and undergraduate students, 3.5 hours, Dubna, Russia.

35) August 2017, Lecture “Around Young tableaux” aimed at high school and undergraduate students, 2 hours, Minsk, Belarus.

36) Fall 2017, Northeastern, MATH 7320, Modern Algebraic Geometry (Invariant theory).

37) July 2018, Lecture series “Catalan numbers: Combinatorics and Algebraic geometry” aimed at high school and undergraduate students, 3.5 hours, Dubna, Russia.

38) December 2018, Lecture series “Introduction to Invariant theory” for undergraduate and master students, HSE, Moscow, Russia. 12 hours.

39) Spring 2019, University of Toronto, MAT 224, Linear Algebra II, 2 sections.

40) Fall 2019, Yale University, MATH 380, Modern Algebra I.

41) Spring 2020, Yale University, MATH 757, D-modules.

42) May 2020, Independent University of Moscow, Introduction to quiver varieties. 20 hours, via Zoom.

43) Fall 2020, Yale University, MATH 380, Modern Algebra I.

44) Fall 2021, Yale University, MATH 380, Modern Algebra I.

45) Spring 2022, Yale University, MATH 603, Introductory topics in Representation theory.

46) June 2022, MIT, a mini-course on quantized symplectic singularities and applications to Lie theory. 14 hours.

47) Fall 2022, Yale University, MATH 720, Topics in Representation theory.

48) Spring 2023, Yale University, MATH 353, Introduction to Representation theory.

49) Fall 2023, Yale University, MATH 380, Algebra.

50) Spring 2025, Yale University, MATH 761, Invariant theory.

51) Fall 2025, Yale University, MATH 3800, Algebra.

52) Spring 2026, Yale University, MATH 6030, Introductory topics in Representation theory.

## 10. CONFERENCES AND WORKSHOPS ORGANIZED

- 1) Representation theory and geometry of symplectic resolutions, May 2015, Northeastern.
- 2) Transformation groups, December 2017, Independent University of Moscow.
- 3) Quantum structures in Algebra and Geometry, August 2019, Northeastern.
- 4) Clay research workshop on Modular representation theory, October 2019, Oxford.
- 5) Conference “On the crossroads of Algebra, Geometry and Physics”, May 2022, Yale.
- 6) Conference “From Representation theory to Mathematical Physics and back”, June 2022, SCGP.



7) Scientific board of “Geometric representation theory and W-algebras”, August-September 2023, ICMS, Edinburgh.

8) Advances in Representation theory 2025, June 2025, Northeastern, Boston (two weeks, the principal organizer for week 2).

## 11. STUDENTS SUPERVISED

### *Graduate:*

Zac Carlini, Yale, starting 2024.

Do Kien Hoang, Yale, 2020-2025.

Mengwei Hu, Yale, 2022- 2026.

Dmytro Matvieievskiy, NEU, 2016-2022.

Seth Shelley-Abrahamson, MIT, jointly with P. Etingof, 2014-2018.

Jose Simental Rodriguez, NEU, 2013-2017.

Boris Tselikhovskiy, NEU, 2015-2020.

Trung Vu, Yale, starting 2022.

Yaochen Wu, Yale, 2019-2024.

Huijun Zhao, NEU, 2013-2018.

### *Master (advised unofficially):*

Vasily Krylov, HSE (Moscow), jointly with M. Finkelberg, 2016-2019.

Dmitry Korb, HSE (Moscow), jointly with M. Finkelberg, 2013-2014.

### *Undergraduate (advised unofficially):*

Daniil Klyuev, SPSU (Saint Petersburg), 2016-2019.

Uladzislau Stazhinsky, BSU (Minsk), 2011-2013.

Aliaksandr Minets, BSU (Minsk), 2009-2011.

Ruslan Maksimau, BSU (Minsk), 2007-2008.

### *High school (advised unofficially):*

Benjamin Li (Christian Heritage High School, Trumbull, CT), 2020-2022.

## 12. LANGUAGES

Russian, English.

## 13. PROFESSIONAL ACTIVITIES

### **Journals:**

Referee for: Annals of Math., Advances in Mathematics, Algebra and Number theory, Duke Math Journal, IMRN, Journal of Algebra, Journal of Lie theory, Selecta Math., Transformation groups, Invent. Math., J. Amer. Math. Soc., Representation theory, Math. Z, Asterisque, Math. Annalen.

A member of the editorial boards for Forum Math Pi and Sigma, Journal of the AMS, Transformation groups, Selecta Mathematica and Journal of Combinatorial Algebra.

### **Department service:**

The director of graduate studies at Yale Math, Fall 2021-Spring 2023, Spring 2025 onward.